

Elizaveta Rebrova

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Research interests

High-dimensional Probability, Non-asymptotic Random Matrix Theory,
Mathematics of Data, Matrix and Tensor Methods, Randomized Algorithms

Employment

2018 - now **University of California, Los Angeles**
Assistant Adjunct Professor
Three year postdoc at the Department of Mathematics
Mentors *Deanna Needell, Terence Tao*

CAM UCLA REU Academic Mentor (Summer 2020)
Worked with 5 undergraduate students on topic modeling for CIP data
PI *Deanna Needell*

2017 - 2018 **Lawrence Berkeley National Lab, Berkeley, CA**
Graduate student Intern (Summer 2017)
Engineer I (Summer 2018)
Computational Research Division, Scalable Solvers Group;
Investigated applications of STRUMPACK linear solver to kernel matrices
Mentor *Xiaoye Sherry Li*

Education

2013 - 2018 **University of Michigan, Ann Arbor**
Ph.D. in Mathematics
Thesis *Spectral properties of heavy-tailed random matrices*
Advisor *Roman Vershynin*

2007 - 2012 **Lomonosov Moscow State University**
Specialist degree in Mathematics (B.S.+M.S. equivalent)
Diploma with honors, 5.00/5.00 GPA
Master thesis *Sobolev classes on infinite dimensional spaces*
Advisor *Vladimir I. Bogachev*

2011 - 2012 **Yandex School of Data Analysis, Moscow**
M.S. equivalent in Computer Science

Publications and preprints

Preprints

(1) M. Iwen^{*1}, D. Needell*, E. Rebrova*, A. Zare*, *Lower Memory Oblivious (Tensor) Subspace Embeddings with Fewer Random Bits: Modewise Methods for Least Squares*, preprint available at arXiv:1912.08294

^{1*} - equal contribution (alphabetical order)

Journal articles

- (7) E. Rebrova, D. Needell, *On block Gaussian sketching for iterative projections*, accepted to **Numer. Algorithms** (2019)
- (6) Y. Liu, W. Sid-Lakhdar, E. Rebrova, P. Ghysels and X.S.Li, *A Parallel Hierarchical Blocked Adaptive Cross Approximation Algorithm*, accepted to **Int. J. High Perform. C.** (2019)
- (5) E. Rebrova, *Constructive local regularization of the operator norm of random matrices*, **J. Theor. Probab.**, Vol. 33(3), pp 1768-1790 (2020)
- (4) E. Rebrova*, R. Vershynin*, *Norms of random matrices: local and global problems*, **Adv. Math.**, Vol. 324, pp 40–83 (2018)
- (3) E. Rebrova*, K. Tikhomirov*, *Coverings of random ellipsoids, and invertibility of matrices with i.i.d. heavy-tailed entries*, **Israel J. Math.**, Vol. 227(2), pp 507-544 (2018)
- (2) V.I. Bogachev*, A.Y. Pilipenko*, E. Rebrova*, *Classes of functions of bounded variation on infinite-dimensional domains*, **Dokl. Math.**, 88(1), pp 391-395 (2013)
- (1) V.I. Bogachev*, E. Rebrova*, *Functions of bounded variation on infinite-dimensional spaces with measures*, **Dokl. Math.**, 87(2), pp 144-147 (2013)

Conference publications

- (5) J. Haddock*, D. Needell*, E. Rebrova*, W. Swartworth*, *Stochastic Gradient Descent Methods for Corrupted Systems of Linear Equations*, Proc. **CISS** (2020)
- (4) G. Chavez, E. Rebrova, Y. Liu, P. Ghysels and X.S.Li, *Scalable and Memory-Efficient Kernel Ridge Regression*, Proc. 34th IEEE **IPDPS** (2020)
- (3) E. Rebrova, D. Needell, *Sketching for Motzkin's iterative method for linear systems*, Proc. 50th IEEE **Asilomar SS&C** (2019)
- (2) E. Rebrova, D. Needell, *New bounds for the block Gaussian sketch and project method*, Proc. **ITA** (2019)
- (1) E. Rebrova, G. Chavez, Y. Liu, P. Ghysels and X.S.Li, *A study of clustering techniques and hierarchical matrix formats for kernel ridge regression*, 32th IEEE **IPDPS ParLearning workshop** (2018)

Fellowships, grants, awards

2018 - 2020	Capital Fund Management sponsored postdoc grant, UCLA
2018	Allen Shields Memorial Fellowship, University of Michigan
2017	NSF ORISE scholarship for summer research internship at LBNL
2017	Rackham One-Term Dissertation Fellowship, University of Michigan
2014 - 2016	University of Michigan Math Department Graduate Summer Fellowships
2013	University of Michigan recruitment bonus
2012	RFFI 10-01-00518, research grant, Russia

Teaching experience

2018 - now	Professor , University of California, Los Angeles • Probability I: Fall 2018 and Spring 2020 (restructured for Math majors) • Statistics: Winter 2019, Spring 2019, Winter 2020
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	• Stochastic Processes: Spring 2020 • Optimization: Fall 2020
2014-2017	Graduate Student Instructor , University of Michigan, Ann Arbor • Calculus I: Fall 2014 (Teacher role in an IBL classroom) • Differential Equations: Winter 2015, Winter 2016, Fall 2016 (Teaching assistant role, including discussion sessions and labs) • Computational Finance: Fall 2017 (Teaching assistant role)
2017	Machine learning instructor , Lawrence Berkeley National Lab, Berkeley • Led 16-session review of the classical machine learning methods
2013-2014	Instructor in Math Lab, University of Michigan, Ann Arbor • One-on-one tutoring on all lower level undergraduate math classes
2012-2013	Algebra instructor , Kolmogorov math and physics high school, Moscow, Russia • Teaching assistant role, including some lecturing and guiding solving algebra problems in class one-on-one with the students
2008-2012	Calculus instructor , Moscow State School 57, Moscow, Russia • Teaching assistant role, guiding solving olympiad-style and calculus problems in class one-on-one with the students

Service and leadership

Mentoring	Mentoring five undergraduate students in CAM UCLA REU program (Summer 2020) topic modeling on text data for California Innocence Project (CIP) PI Deanna Needell, assistant mentor Denali Molitor Co-leading three student projects for the UCLA Math 290J class (Spring 2020) (1) topic-based search for COVID-19 related scientific literature (2) topic modeling for COVID-19 related Twitter data (3) studying of the influence of demographics factors to the spread of COVID-19 Co-mentoring PhD student Will Swartworth (2019 - now) scientific advisor Deanna Needell, co-mentor Jamie Haddock
Leadership	Co-organizing Reading Random Matrices Seminar (2018 - 2019) University of California, Los Angeles, CA faculty contact Terence Tao, co-organizer Palina Salanevich Co-organizing Student Analysis seminar (2015 - 2016) University of Michigan, Ann Arbor, MI co-organizer Siddhant Agrawal
Events	Member of Organizing Committee (2020) /postponed/ Southern California Applied Math Symposium (SoCAMS) Member of Organizing Committee (2007 - 2012) Mathematical and Linguistics Olympiads for high school students Moscow, Russia
Reviewing	Linear Algebra and Applications

Invited talks

June 2020	Online Asymptotic Geometric Analysis Seminar
April 2020	Graduate Seminar California State University Channel Islands, CA
November 2019	Combinatorics and Probability Seminar University of California, Irvine, CA
November 2019	Asilomar Conference on Signals, Systems, and Computers Pacific Grove, CA
October 2019	Probability Seminar Stanford University, Palo Alto, CA
October 2019	Probability Seminar University of Southern California, Los Angeles, CA
March 2019	High-dimensional Seminar Georgia Institute of Technology, Atlanta, GA
March 2019	Probability Seminar University of Alberta, Edmonton, Canada
February 2019	Southern California Applied Mathematics Symposium (SoCAMS) California Institute of Technology, Pasadena, CA
September 2018	Structural Inference in High-Dimensional Models worksop, National Research University Higher School of Economics, Moscow, Russia
August 2018	Summer Informal Regional Functional Analysis Seminar (SUMIFRAS) Texas A&M University, College Station, TX
June 2017	Probability Seminar Université Paris Diderot, Paris, France
June 2017	Probability Seminar Université-Paris-Est Marne-la-Vallée, Champs-sur-Marne, France
May 2017	Analysis Seminar University of Alberta, Edmonton, Canada
March 2017	MIC Seminar Center for Data Science, NYU, New York
June 2016	CMS Summer Sectional Meeting University of Alberta, Edmonton, Canada
March 2016	AMS Sectional Meeting University of Georgia, Athens, GA

Other talks

2019	6 expository talks at Reading Random Matrices Seminar University of California, Los Angeles, CA
2018	Research talk at Random Matrices Seminar University of California, Los Angeles, CA
2015 - 2017	3 expository talks at Analysis/Probability Learning Seminar University of Michigan, Ann Arbor, MI

2014 - 2015	3 expository talks at Student Analysis Seminar University of Michigan, Ann Arbor, MI
2014 - 2015	Series expository of talks on Geometric Analysis Reading Seminar University of Michigan, Ann Arbor, MI

Conferences and workshops participation

November 2019	Asilomar Conference on Signals, Systems, and Computers Pacific Grove, CA
February 2019	Southern California Applied Mathematics Symposium (SoCAMS) California Institute of Technology, Pasadena, CA
August 2018	Summer Informal Regional Functional Analysis Seminar (SUMIFRAS) Workshop Texas A&M University, College Station, TX
May 2018	Random Matrices and Free Probability Theory Workshop III within Quantitative Linear Algebra long program Institute for Pure and Applied, Los Angeles, CA
November 2017	Geometric Functional Analysis and Applications Workshop Mathematical Sciences Research Institute, Berkeley, CA
August 2017	Phenomena in High Dimensions Introductory Workshop within Geometric Functional Analysis and Applications long program Mathematical Sciences Research Institute, Berkeley, CA
August 2017	Geometry and Probability in High Dimensions, Connections for Women Workshop Mathematical Sciences Research Institute, Berkeley, CA
July 2016	The Mathematics of Data, Graduate Summer School IAS/Park City Mathematics Institute, Park City, UT
June 2016	CMS Summer Meeting University of Edmonton, Alberta, Canada
June 2016	Summer School on Random Matrices University of Michigan, Ann Arbor, MI
April 2016	Informal Analysis Seminar Kent State University, Kent, OH
March 2016	AMS Sectional Meeting University of Georgia, Athens, GA
November 2015	Informal Analysis Seminar Kent State University, Kent, OH
November 2015	Analytic and Probabilistic Techniques in Modern Convex Geometry Conference University of Missouri, Columbia, MO
April 2015	Workshop on Information Theory and Concentration Phenomena Institute for Mathematics and its Applications, Twin Cities, MN
March 2015	Informal Analysis Seminar Kent State University, Kent, OH
October 2014	Informal Analysis Seminar University of Michigan, Ann Arbor, MI
May 2014	Random Matrix Theory School, program for Women and Mathematics Institute for Advanced Study, Princeton, NJ

May 2012	The XIX International Student, Postgraduate and Young Scientist Conference “Lomonosov” Moscow State University, Moscow, Russia
July 2006 - 2009	Summer School “Contemporary Mathematics” Ratmino, Russia

Technical skills

Math Software	Matlab, Excel/Sheets, L ^A T _E X
Languages	Python, C/C++
ML related	jupyter, pandas, numpy, scikit learn
Other	git, shell

Languages	English (fluent), Russian (native), German (basic), French (basic)
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